EmailCheck — Email Verification Web Application

Project Submission Document | Senior Full-Stack Engineering Deliverable

Submission Date: October 25, 2026	**Stack:** Python 3.11, Flask 3.1, dnspython, Vanilla JS, CSS3
Test Suite: 31 / 31 Tests Passed (100% Pass Rate)	**Production WSGI:** Gunicorn (Render / Cloud Native Ready)

1. Executive Project Overview

EmailCheck is a responsive web application built to analyze email addresses and estimate deliverability without transmitting messages. The system inspects email syntax according to RFC 5322 specifications, performs DNS resolution for Mail Exchanger (MX) routing including RFC 5321 implicit MX fallback, and safely probes mail servers via SMTP handshake when available. Addresses are conservatively classified as **Likely to be Delivered**, **Likely to Bounce**, or **Unknown**.

2. Application URLs & Repository Status

Item	Status	URL / Action Details
GitHub Repository	**Verified & Active**	Active Public GitHub Repository: https://github.com/Zenith150/emailcheck
Live Cloud Deployment	**Pending Deployment**	Configured for Render deployment via render.yaml and Procfile. Health check configured at /health. Target URL: https://emailcheck.onrender.com *(Pending Render web service link)*
Local Server Verified	**100% Verified**	Verified live on local host (http://127.0.0.1:5000). Health check returns 200 OK. API verified with live domains (Google, Cloudflare, invalid syntax, NXDOMAIN).

3. Transparent Classification Framework

Classification	Criteria & Non-Invasive Diagnostics	Explanation Provided to User
Likely to be Delivered	Valid RFC syntax; verified domain with active MX records or implicit MX (A/AAAA fallback). Non-invasive checks find no delivery defects.	Valid syntax and active MX routing verified. Note: Delivery is estimated based on mail server routing; mailbox delivery is not guaranteed.
Likely to Bounce	Clearly malformed syntax, nonexistent domain (NXDOMAIN), domain without mail routing (no MX/A records), or permanent SMTP rejection (550).	Explicit diagnostic reason provided (e.g., malformed syntax, domain not found in DNS, or mailbox explicitly rejected by mail server).
Unknown	DNS timeout/server error, cloud host outbound SMTP port 25 blocked, server refuses verification (252), greylisting (4xx), or catch-all server.	Explains exact limitation (e.g., DNS timeout, catch-all configuration preventing individual mailbox check, or port 25 restricted on host).

4. Verification Limitations & Hosting Compatibility

Inherent Limitations of Verification: Email verification cannot guarantee inbox placement. Mailbox existence checks without sending an email are fundamentally constrained because modern mail servers frequently employ spam defenses such as greylisting, tarpitting, silent drop, or catch-all routing. Furthermore, free cloud hosting platforms (such as Render) intentionally restrict outbound TCP port 25 to protect reputation and mitigate spam abuse.

EmailCheck Safe Handling: The application detects host port 25 capabilities dynamically. When port 25 is blocked, EmailCheck never fabricates results; it maintains 100% operational functionality for RFC syntax and DNS/MX routing, and conservatively reports **Unknown** for unconfirmable recipient states.

5. Application Architecture & Implemented Features

Component	Implementation Details
User Interface	Modern responsive layout, real-time address counter (max 100), loading indicator, 4 summary metric cards, interactive filter tabs, live table search, RFC 4180 CSV export, and accessible semantic markup.
Verification Engine	verifier.py provides RFC 5322 validation, dnspython MX resolution with public DNS fallback (8.8.8.8/1.1.1.1), implicit MX fallback, safe SMTP probe, catch-all detection, and in-flight memoization.
REST API	POST /api/verify (accepts batch JSON list), GET /health (monitoring & capability probe), sliding-window IP rate limiter (30 req/min), and structured error handlers (400, 404, 429, 500).

Privacy & Security	Zero email transmission (no DATA command), zero database persistence (in-memory processing only), anonymized logging, strict HTML escaping against XSS, and 100 address batch boundary.
Cloud Deployment	Production render.yaml blueprint, Procfile configured for Gunicorn WSGI, dynamic port binding ($PORT), and production dependency manifest.

6. Automated Test Suite Summary

The automated test suite in tests/test_verifier.py and tests/test_api.py was executed and passed completely with **31 passed tests in 0.93 seconds**. External DNS and SMTP networks are mocked via unittest.mock for isolated, deterministic execution:

• Valid and malformed syntax rules (local part length, consecutive dots, missing symbols): **PASSED**
• Domain DNS MX resolution, NXDOMAIN detection, implicit MX A/AAAA fallback, and DNS timeouts: **PASSED**
• Safe SMTP handshakes (250 OK, catch-all probe, 550 permanent reject, 451 greylist, connection refused): **PASSED**
• Batch memoization (caching duplicate lookups within a batch), limits enforcement, and API errors: **PASSED**

Privacy & Non-Invasive Safety Statement: EmailCheck guarantees that no emails are sent at any time. Verification relies solely on non-invasive structural diagnostics. User data is not permanently stored or shared.